

GLA UNIVERSITY

Department of Computer Engineering & Applications

MINI PROJECT – 1 SYNOPSIS

1. BASIC PROJECT DETAILS

Project Title	Subscription Usage & Billing Command Console
Batch No.	2
Section	AE
Semester	5th
Date of Submission	18 September 2026
Faculty Mentor / Guide	Satyam Kumar Jha

2. TEAM MEMBERS & ROLES

S.No.	Name of Student	University Roll No.	Role in Project
1	ADITYA MAHESHWARI	2415000101	Frontend Development & React Component Integration
2	ADITYA SINGH RATHAUR	2415000118	Project Lead, Billing Logic, Usage Calculations & State Management
3	AJAY SINGH	2415000137	UI/UX Design, Responsive Layout & Dashboard Visualization
4	AKASH DUBEY	2415000146	Functional Testing, Edge-Case Validation & Bug Tracking
5	AMOLAK TALIWAL	2415000195	Documentation, Research, Presentation & Project Demonstration

3. PROJECT DESCRIPTION

The Subscription Usage & Billing Command Console is a frontend-only React application designed to provide a centralized interface for visualizing tenant usage, subscription plan tiers, and simulated billing information. The project focuses on demonstrating how a modern subscription management console can work entirely with data generated and stored in the browser, without using any backend server, external database, or real payment gateway.

The application will display tenant-wise resource usage against defined plan limits through interactive meters that animate in real time. It will provide simulated subscription plans and allow an administrator to perform plan upgrades, downgrades, or cancellations. Whenever a plan change is made, the system will immediately recalculate the simulated invoice, including plan charges, applicable prorations, credits, and the updated billing total.

A dunning-style notification center will also be included to surface simulated failed-payment states. These notifications will display retry information and countdown timers implemented purely on the client side. All application data, state changes, calculations, and timers will be handled through React state and browser storage. The project will demonstrate component-based frontend development, responsive dashboard design, state management, client-side business logic, data visualization, and functional testing.

4. OBJECTIVES & PROPOSED WORK (WHAT WE ARE GOING TO DO)

- To develop a frontend-only subscription usage and billing command console using React.
- To generate and store simulated tenant, plan, usage, invoice, and payment-status data in the browser.
- To visualize tenant resource usage against subscription limits using responsive and animated meters.
- To provide multiple subscription plan tiers with clearly defined limits and pricing information.
- To implement simulated plan upgrade, downgrade, and cancellation workflows for an administrator.
- To calculate and display invoice changes, including prorations, credits, and revised billing totals.
- To create a dunning-style notification center for simulated failed-payment states.
- To implement client-side retry countdown timers without any server-side dependency.
- To use reusable React components, controlled state, and modular frontend structure.
- To validate user interactions and billing calculations through functional and manual UI testing.
- To maintain a responsive and user-friendly interface suitable for desktop and mobile screens.

5. FUTURE SCOPE / ENHANCEMENTS

- Integration with a backend service and a real database for persistent multi-tenant data.
- Integration with real payment gateways and payment-status webhooks.
- Authentication and authorization for administrators, billing managers, and customers.
- Automated recurring billing, subscription renewal, and invoice delivery.
- Email and SMS notifications for failed payments, upcoming renewals, and usage limits.
- Exporting invoices in PDF or other standard formats.
- Advanced usage analytics, reports, and historical billing charts.
- Cloud deployment and scalable data storage.
- Integration with external subscription and payment APIs.
- Audit logs and role-based access control for production environments.

6. TOOLS & TECHNOLOGIES USED

Category	Technology / Tool
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Frontend Framework	React.js
Programming Language	JavaScript / JSX
Styling & UI	HTML5, CSS3, Responsive Web Design
State & Browser Data	React State, Browser Storage / LocalStorage
Visualization	React-based usage meters, progress indicators and dashboard components
Version Control	GitHub
Development Environment	Visual Studio Code, Node.js and npm
Testing	Manual UI Testing, Functional Testing and Edge-Case Validation

7. EXPECTED OUTCOME

The expected outcome of the project is a functional, responsive, and interactive Subscription Usage & Billing Command Console developed entirely with React. The application will allow an administrator to view simulated tenant usage, understand plan tiers, monitor usage against limits, and simulate subscription lifecycle actions such as upgrade, downgrade, and cancellation.

The system will immediately reflect plan changes in the simulated invoice breakdown by applying client-side billing logic and prorations. It will also provide a notification center for simulated failed-payment situations with retry countdowns rendered through browser-based timers. The completed project will demonstrate practical knowledge of React components, state management, browser storage, responsive UI design, client-side calculations, visualization, validation, and testing.

The project will serve as a frontend prototype and a foundation for future expansion into a complete subscription management and billing platform with backend services, authentication, real payment processing, and cloud deployment.

Signature of Faculty Mentor